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On Sampling Invariant Signal Representation

Or: On the effect of sampling on algebraic relations

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Abstract

Most signal processing systems are based on discrete-time signals although the origin of many sources of information is analog. In this work we consider the task of signal representation by a set of basis functions. Presently, without prior knowledge of the signal beyond its samples, no bound on the potential representation error is available. The question raised in this paper is to what extent the sampling process keeps algebraic relations, such as inner product, intact. By interpreting the sampling process as a linear bounded operator, an upper bound on the representation error is derived and demonstrated. Based on our theorems, one can then determine the maximum representation error induced by the sampling process. We further propose a new approximation scheme for the calculation of the inner product, which is optimal in the sense of minimizing the maximum representation error. Our results are applicable to signal processing systems where analog signals are represented by their sampled versions.